Phonologically conditioned affix order in Yidiɲ: Evidence for strata

Marie-Luise Popp

November, 2024

Abstract

This paper discusses the non-uniform behavior of different affixes in phonologically conditioned affix order in Yidiɲ. I present evidence from final syllable deletion illustrating that the word comprises two different cyclic domains. Phonologically conditioned affix order is bound to the first cyclic domain, hence being counter-fed by affixes of the second layer. This study further substantiates the model of the morphology-phonology interface as proposed by Stratal Optimality Theory, which posits temporary phonological access to morpheme boundaries, predicting phonological conditioned affix order to operate within one stratum.

1 Introduction

In this paper, I provide a unified analysis of two morphophonological phenomena in Yidiɲ (Pama-Nyungan, Australia). Most importantly, different affixes exhibit non-uniform behaviour in these processes. Some affixes in Yidiɲ have the property of lengthening their preceding vowel. The relative order of the prelengthening aspectual affixes GOING/COMING and the COM/CAUS suffix *ŋa* is driven by the number of syllables of the root: the GOING suffix precedes the COM/CAUS suffix after disyllabic roots in (1-a), but follows the COM/CAUS suffix after trisyllabic roots in (1-b). However, the prelengthening INTRANS affix *:ɖi* strictly precedes the inflectional PURP affix *na* after trisyllabic roots in (1-c) despite *na* being strikingly similar in form to COM/CAUS suffix *ŋa*.

(1) Phonologically conditioned affix order in Yidiɲ (Dixon 1977: 228)

a. magi-:ri-ŋa-l-ŋ
   [ma.gi:.ɻi.ŋal]
   climb.up-GOING-COM/CAUS-CL-PRS

b. maɖinda-ŋa-:li-ŋ
   [ma.ɖin.da.ŋa:.liŋ]
   walk.up-COM/CAUS-GOING-PRS

c. wuŋaba-:ɖi-na
   [wu.ŋa.ba.ɖi:.na]
   hunt.for-INTRANS-PURP

In Yidiɲ, most morphemes lose their vowels in final position if this deletion process creates words with an even number of syllables. This is illustrated in (2-a), where the COM/CAUS *ŋa* loses its vowel after a disyllabic root, since the word would otherwise have an odd number of syllables. Again, different affix groups exhibit asymmetries in final syllable deletion. Prelengthening suffixes, such as GOING *:li* never lose their vowel, see (2-b). The COM/CAUS *ŋa* shortens but resists shortening if deletion of the vowel is ruled out by phonotactic reasons: in (2-c), deletion of the final vowel would create *(gu.ri:ɲŋ)*, which has an illicit coda cluster. With inflectional affixes, such as *ɲunda* in (2-d), consonants may be deleted along with the vowel if deletion would otherwise be prohibited by phonotactics. In (2-d), the entire last syllable *da* is deleted since deletion of the vowel only would create the form *(bi.nar).(ŋal.ɲu:nd)*, which again contains an illicit coda.

(2) Final syllable deletion in Yidiɲ (Dixon 1977: 55, 349, 478)

a. /gada-ŋa/
[gada:ŋ]
come-COM/CAUS

b. wawa-:li
[wa.wa:.li]
see-GOING

c. /guriɲ-ŋa/
[gu.ri:ɲ.ŋa]
'good-COM/CAUS'

d. /binarŋa-l-ɲunda/
[bi.nar.ŋal.ɲu:n]
'warn-CL-DAT.SUBORD'

In this paper, I suggest that the differences between the affixes can be explained by two core assumptions. Firstly, I propose that prelengthening affixes come with underlying foot structure. Feet can never be altered in Yidiɲ, thereby resisting final syllable deletion. This representational difference also accounts for the phonologically conditioned affix order in (1). In section 4, I illustrate that the examples in (1) instantiate a case of morpheme reordering in the phonological component of the grammar, caused by universal principles regarding the wellformedness of foot structure, such as foot binarity. It has previously been doubted most prominently by Paster (2009) that morpheme reordering in the phonology exists. However, Stratal Optimality Theory (StratOT) (Kiparsky 2000, Bermúdez-Otero 2011, 2018) predicts the existence of phonologically conditioned affix order, as phonology has access to morpheme boundaries within one stratum. I will argue that phonologically conditioned affix order in Yidiɲ patterns exactly as predicted by StratOT. Crucially, it applies only within the first cyclic domain of the word, which includes derivational affixes but excludes inflectional affixes. As a consequence, the affixes in (1-c) cannot be reordered for phonological reasons because the inflectional affix *na* is only affixed at the second cyclic domain, thus counter-feeding PCAO. Evidence for inflectional affixes attaching later comes from the asymmetry between inflectional affixes and the COM/CAUS affix in final syllable deletion. As mentioned above, inflectional affixes may lose their consonants to produce licit codae while this option is blocked with the COM/CAUS affix, compare (2-c) vs. (2-d). I argue that this asymmetry indicates that the COM/CAUS affix has already been syllabified when final syllable deletion applies, suggesting that it already underwent a cycle of phonology. In sum, the analysis I put forward in this paper does not only provide a unified analysis of phonologically conditioned affix order and final syllable deletion in Yidiɲ, but reinforces the idea of the morphology-phonology interface in exactly the form suggested by Bermúdez-Otero (2018). Specifically, the data in Yidiɲ are compelling evidence that the word consists of exactly two word-internal strata, in which the phonological component of the grammar has temporary access to morphological structure.

This paper is structured as follows. I introduce the general morphological makeup including morphological and phonological properties of different affix groups in section 2. In section 3, I discuss the phenomenon of final syllable deletion and the asymmetries of affixes with respect to this phenomenon. I propose a formal perspective on PCAO assuming that prelengthening affixes are born with foot structure in section 4. In section 5, I demonstrate how final syllable deletion can be accounted for under the assumption that the word in Yidiɲ consists of exactly two cyclic domains. Both assumptions lend themselves to an analysis in Stratal Optimality Theory, which I discuss in 6. Section 7 contains a discussion of remaining issues before I draw a conclusion in section 8.

2 The phonology and morphology of suffixes in Yidiɲ

Verbs in Yidiɲ are entirely suffixing. An overview of the morphological structure is given in Table 1. Dixon (1977) distinguishes three different groups of suffixes, depending on their morphological and phonological properties.

Slot 1	**Slot 2**	**Slot 3**	**Slot 4**	**Slot 5**	**Slot 6**		**Slot 7**
root	derivational affixes (-n/-l/-ɻ)			class marker	inflection		clitics
	COM/CAUS		*-ŋa/-maŋa/-maŋa -l*	*-n/-l/-ɻ*	PST	*-ɲu*	NOW -(a)la
	INTRANS		*-ːɖi -n*		NPST	*-ŋ/-l/-ɻ*	SELF *-di*
	GOING		*-ŋali/-ːli/-ːɻi -n*		IMP	–[-APPROX]	
	COMING		*-ŋada/-ːda/-ːda -n*		PURP	*-na*	
					DAT.SUBORD	*-ɲunda*	
					CAUS.SUBORD	*-ɲum*	
					SUBORD	*-ɖi*	

Table 1: Affixes in Yidiɲ (based on Dixon 1977: 210)

Each verbal form may take up to three *derivational affixes*, which have been allocated to slots 2–4 in Table 1. The derivational affixes include the valency-increasing suffix COM/CAUS, the suffix INTRANS, as well as two aspectual affixes GOING and COMING. The valency-increasing suffix COM/CAUS has two core functions. It has the function of a comitative applicative. As such, it has two syntactic effects: first, it turns an oblique argument into a core argument, like *buɲa* 'woman' in (3), which carries comitative case in the absence of the COM/CAUS verbal suffix in (3-a), but zero-marked ABS case in the presence of the COM/CAUS *-ŋa* in (3-b). Second, the predicate becomes transitive, thus the agent argument bears absolutive case in the absence of a COM/CAUS in (3-a), but ergative case when the verb comes with the COM/CAUS suffix in (3-b).

(3) COM/CAUS as a comitative applicative (Dixon 1977: 293)

a. Waguːɖa-∅ buɲa-ːy gali-ŋ.
   man-ABS woman-COM/CAUS walk-NPST

b. Waguːɖa-ŋgu buɲa- gali:-ŋa-l
   man-ERG woman-ABS walk-COM/CAUS-CL:L
   'The man is going with the woman.'

As a causative marker, the COM/CAUS turns the sole argument of an intransitive verb into an internal argument of a transitive verb. Hence, the NOM-marked 1SG pronoun *ŋayu* in

(4-a) appears in ACC case in the presence of the COM/CAUS suffix in (4-b). Additionally, the COM/CAUS suffix introduces a new external argument carrying ERG case.

(4) COM/CAUS as a causative applicative (Dixon 1977: 312)

a. ŋayu warŋgi:-ɲ.
   1SG.NOM turn.around-PST
   'I turned around.'

b. ŋaɲaɲ guɖunu-ŋgu warŋgi-ŋa-l-ɲu.
   1SG.ACC wind-ERG turn.around-COM/CAUS-CL:L-PST
   'The wind turned me around.

The suffix INTRANS *-:ɖi* has variety of functions. It may function as a valency-decreasing reflexive or antipassive marker. In these functions, it combines with transitive predicates and produces intransitive predicates. It may also appear with a transitive root to indicate that the agent argument lacks volitional control over the action, though in this case, the predicate remains transitive. Additionally, it can be applied to both transitive and intransitive verbs, maintaining the original valency, to indicate that the action is either an ongoing and/or continuous action. An example of the INTRANS suffix in its antipassive function is given in (5). Crucially, the INTRANS suffix turns a transitive root with both an ergative and an absolutive argument in (5-a) into an intransitive predicate with the agent argument marked with ABS case and the theme argument marked with OBL case in (5-b).

(5) INTRANS as an antipassive

a. Waguɖa-ŋgu guda:ga-∅ wawa:-l.
   man-ERG dog-ABS see-CL:L

b. Waguɖa-∅ wawa:-ɖi-ɲu gudaga-nda
   man-ABS see-INTRANS-PST dog-DAT
   'The man saw the dog.'

All combinations of derivational affixes are attested. However, the aspectual derivational affixes are mutually exclusive, as the former marks direction towards the center of speech, while the latter markers direction away from the center of speech. Table 1 further shows that some of the derivational affixes have different allomorphs. The choice of the allomorph is not phonologically conditioned. Rather, the surface form depends on the verb class of the base. In Yidiɲ, each verb root belongs to one of three inflection classes. The inflection classes are labeled *n, l* and *ɻ* based on the final consonant that they entail. The class markers are glossed as CL:N, CL:L and CL:R, respectively and may be omitted in certain phonological environments. A property that distinguishes derivational affixes from other suffixes is the fact that the derivational suffix also entails a class marker, which is spelled out in slot 5 in Table 1. Specifically, the COM/CAUS brings the class marker *-l*, while all other derivational affixes entail the class marker *-n*. Inflectional affixes, on the other hand, do not have class-dependent allomorphs. Moreover, inflectional affixes are obligatory and mutually exclusive, while derivational affixes are optional. In short, each verbal form in Yidiɲ will occur with exactly one inflectional affix and 0–3 derivational affixes. Dixon (1977) further lists clitic-like suffixes, which are given in slot 7 in Table 1. I will not discuss the clitic-like suffix in considerable detail here, since they do not participate in the phonological processes that are relevant for discussion.
Table 1 further reveals that some allomorphs of derivational affixes have the property of lengthening the preceding syllable. The examples in (6-a,b) show that the INTRANS suffix

lengthens the preceding syllable, independent of the syllable number of the root. For the GOING and the COMING suffixes, only the *l* and *ɻ* class allomorphs lengthen the preceding syllable, as seen in (6-c). Note also that there is no trisyllabic root of class *ɻ* and only one trisyllabic root of class *l*, which seems incompatible with aspectual suffixes.

(6) Vowel length through affixation (Dixon 1977: 218)

a. wuŋaba-:ɖi
   [wu.ŋa.baː.ɖi]
   hunt.for-INTRANS
b. buga-:ɖi
   [bu.gaː.ɖi]
   eat-INTRANS
c. wawa-:li
   [wa.waː.li]
   see-GOING

Dixon (1977) observes that the relative order of prelengthening allomorphs of GOING and COMING and the COM/CAUS suffix depends on the number of syllables of the root. In (7-a), the root is the bisyllabic, *l* class root *magi*. In this case, the *l* class allomorph of the GOING affix is selected, which is the prelengthening suffix *-:ɻi*. In combination with the monosyllabic COM/CAUS affix *ŋa*, the GOING suffix precedes the COM/CAUS, thus lengthening the second syllable of the root. In (7-b), however, the root is trisyllabic. In this case, the GOING suffix follows the COM/CAUS and length goes to the COM/CAUS affix and hence, the fourth syllable of the word.[1]

(7) Placement of the GOING and COM/CAUS suffix (Dixon 1977: 228)

a. magi-:ri-ŋa-l-ŋ
   [ma.giː.ɻi.ŋal)]
   climb.up-GOING-COM/CAUS-CL-PRS
b. maɖinda-ŋa-:li-ŋ
   [ma.ɖin.da.ŋaː.liŋ]
   walk.up-COM/CAUS-GOING-PRS
c. *[ma.ɖin.daː.li.ŋaŋ]

Dixon (1977) makes the following empirical generalization: the COM/CAUS and the prelengthening aspectual suffixes always surface in the order that makes the inherent length of the aspectual allomorph surface on an even-numbered syllable. Hence, the relative order of affixes in Yidiɲ is an instance of phonologically conditioned affix order (henceforth PCAO).

When the prelengthening suffixes occur with a monosyllabic, inflectional affix, however, reordering does not take place, as shown in (8). From the data in (7), we might expect the inflectional PURP affix *na* to precede the INTRANS suffix, as in (8-b). In this case, the inherent length would occur on the second syllable of the second foot, and hence, on an even-numbered syllable. However, reordering does not seem to be an option: the INTRANS suffix precedes the PURP suffix. The example in (8-a) shows that the inherent length of the INTRANS suffix does not fall onto the preceding syllable. Rather, the INTRANS suffix itself is lengthened.

[1] Note also that Dixon (1977) lists these forms without providing a translation or any other clue about which order might be default or underlying.

(8) Relative order of INTRANS *-:ɖi* and PURP *na*

a. wuŋaba-:ɖi-na
[(wu.ŋa).(ba.ɖi:).na]
hunt.for-INTRANS-PURP

b. *[(wu.ŋa).(ba.na:).ɖi]

c. *[(wu.ŋa).(ba:.ɖi).na]

From this section, we can conclude that there are three types of affixes. First, there are three different monosyllabic and prelengthening derivational suffixes. Second, there is the monosyllabic derivational suffix *ŋa*, whose relative position to the prelengthening aspectual suffixes is driven by phonological factors. Third, there are monosyllabic, inflectional suffixes, which always follow derivational suffixes, independent of the phonological structure of the root. In the following, I will delve into another phonological process in Yidiɲ, which also exhibits asymmetries between these three types of affixes.

3 Affix asymmetries in final syllable shortening

Yidiɲ has another phonological process that makes reference to the number of syllables. The vowel of the final syllable is deleted if non-deletion would result in a word with an odd number of syllables. However, final deletion does not occur with all morphemes. In general, final syllable deletion follows the rule in (9).

(9) Final syllable deletion
CV(C).CV(C).CV ⇒ CV(C).CV:(C)C

If a word ends up with an odd-number of syllables, the vowel of the final syllable is deleted, the onset of the final syllable becomes the coda of the preceding consonant, and the vowel of the preceding syllable is lengthened. A simple example of final syllable shortening is illustrated in (10). In (10-a), the trisyllabic root *gindanu* combines with the ergative case marker *ŋgu*. The resulting complex word has four syllables, which fall into two binary feet. In absolutive contexts, however, the root does not carry any affixes, and remains trisyllabic. The final vowel is deleted, and the onset of the final syllable */n/* becomes the coda of the preceding syllable, as in (10-b).

(10) Word-final deletion (Dixon 1977: 57)

a. /gindanu-ŋgu/
[(gin.da)(nu.ŋgu)]
'moon-ERG'

b. /gindanu/
[(gin.da:n)]
'moon.ABS'

Final syllable deletion affects both roots and suffixes, with exceptional non-participating morphemes in both groups. As an example, all prelengthening suffixes resist final syllable shortening, as shown in (11). Although the combinations of prelengthening suffixes and bisyllabic roots yield words with three syllables in total, the final vowel is not deleted, see (11-b,d). While the ungrammaticality of (11-b) could be explained by the fact that /ɖ/ is not a licit coda, the ungrammaticality of (11-d) is unexpected. Dixon (1977) concludes that the participation in final syllable shortening is an idiosyncratic property of the affixes.

(11) No final shortening with prelengthening affixes (Dixon 1977: 218)

a. buga-:ḍi
   [(bu.ga:).ḍi]
   eat-INTRANS
b. *[(bu.ga:ḍ)]
c. wawa-:li
   [(wa.wa:).li]
   see-GOING
d. *[(wa.wa:l)]

In general, final syllable deletion only applies when shortening produces licit codae. This is illustrated in (12). The COM/CAUS shortens, as in (12-a). After a nasal-final root, however, deletion of the vowel would produce the illicit coda combination /*ɲŋ*/. It is not possible to delete the suffixes altogether, as in (12-e). The same observation holds for roots, see (12-f). The root *binarŋa* does not shorten to *bina:rŋ*, since /*rŋ*/ is not a licit coda. Again, it is not possible to delete consonantal root material to produce licit codae, either, as in (12-h).

(12) Final shortening with roots and COM/CAUS (Dixon 1977: 55, 349, 478)

a. /gada-ŋa/
   [gada:ŋ]
   come-COM/CAUS
b. *[[gadaŋa]]
c. /guriɲ-ŋa/
   [(gu.ri:ɲ)ŋa]
   'good-COM/CAUS'
d. *[(gu.ri:ɲŋ)]
e. *[(gu.ri:ɲ)]
f. /binarŋa/
   [bina:rŋa]
   'warn'
g. *[bina:rŋ]
h. *[bina:r]

In the previous section, we have seen that monosyllabic derivational affixes behave differently in PCAO than monosyllabic inflectional affixes. Another asymmetry between the two groups arises in final syllable shortening. The examples in (12) have demonstrated that the COM/CAUS suffix will only shorten if the resulting word ends in a licit coda. The same generalization holds for inflectional affixes. However, consonants of inflectional affixes may be deleted to produce licit codae. This is illustrated in (13). The example in (13-a) involves the inflectional affix *ɲunda*. Final syllable deletion would result in the form in (13-b), which involves an illicit coda /*nd*/. As a consequence, a consonant is deleted, resulting in a word with two binary feet and a licit coda. This process might even cause the deletion of entire affixes, as in (13-d). In sum, final syllable deletion reveals another asymmetry between three groups of affixes: prelengthening suffixes never shorten. As for the COM/CAUS suffix, the final vowel may be deleted to produce words with binary feet if the resulting word contains a licit coda. With inflectional syllables, however, both vowel and consonant may deleted.

(13) Final syllable shortening in inflectional affixes (Dixon 1977: 277)

a. /binarŋa-l-ɲunda/
[(bi.nar)(ŋalɲu:n)]
'warn-CL-DAT.SUBORD'

b. *[(bi.nar)(ŋalɲu:nd)]

c. *[(bi.nar)(ŋalɲu:n).da)]

d. /wawa-l-ɲu/
[wawa:l]
'see-CL:L-PST'

In the following section, I will lay out analyses for both phenomena that capture the asymmetries between the affixes. The core idea of the analyses is that the prelengthening affixes have inherent foot structure that protect them from shortening. The difference between the COM/CAUS on the one hand, and the inflectional suffixes on the other hand, arises from the stratal structure of the word in Yidiɲ: the COM/CAUS enters the word at the stem-level, which is the domain for PCAO and syllabification. Inflectional affixes, however, are only concatenated at word-level, thus counter-feeding PCAO.

4 A formal analysis of PCAO in Yidiɲ

Final syllable deletion in Yidiɲ and the exceptional behaviour of certain roots and affixes have already been discussed by Round (2017). Some non-participating morphemes can be predicted from the phonological shape, e.g. bisyllabic affixes do never participate in final syllable deletion. The only exception to this generalization is the inflectional affix *ɲunda*, which is actually a combination of the two monosyllabic affixes *ɲu* and *nda.* Roots where shortening would result in an illicit coda, do not shorten either. In contrast, other non-undergoers cannot be predicted from the phonological or semantic properties (Dixon 1977). This becomes clear from the example in (14). The root *gaɟara* 'possum' and the root *guɟara* 'broom' have a similar underlying form, however, the former shortens (see (14-a)) while the latter does not (see (14-b)). The contrast between the two examples further reveals that there is no phonological explanation for the non-shortening behaviour of *guɟara*: (14-a) shows that the resulting coda is allowed.

(14) Shortening vs. non-shortening roots (Round 2017: 60)

a. /gaɟara/
[(ga.ɟa:r)]
'possum.ABS'

b. /guɟara/
[(gu.ɟa:).ra]
'broom.ABS'

I assume that bisyllabic affixes, which do not participate in final syllable shortening, come with inherent foot structure that prevents shortening. The prelengthening aspectual suffixes *:li* 'GOING' and *:da* 'COMING' have most likely evolved from the independent verb roots *gali* 'to come' and *gada* 'to come' (Dixon 1977). Hence, it seems more than reasonable to assume that prelengthening suffixes resist shortening because they are stored as (now unary) feet in the lexicon. Under these assumptions, the patterns of PCAO in (7) follow neatly from conditions on the wellformedness of feet. Assuming that the GOING suffix is a foot on its own, while the shortening COM/CAUS suffix is not, the GOING suffix may either precede or follow the COM/CAUS suffix, compare (15-a) and (15-b). In order to explain why (15-a) is the observed surface form, we can simply conclude that (15-b) is sub-optimal because the unfooted COM/CAUS suffix is between the left edge of the word and the footed GOING suffix. In short, the feet in (15-b) are not fully left-aligned. Let us now turn to cases with trisyllabic roots. If the GOING suffix would again follow the root, as in (15-d), the unfooted COM/CAUS suffix would appear to its right. Additionally, the trisyllabic root can be assumed to break into a bisyllabic foot with its third syllable unparsed. Again, an unparsed syllable would intervene between the footed GOING suffix and the left edge of the word. Reversing the order, however, improves the situation, illustrated in (15-c). In this order, the third syllable of the root forms a binary foot with the otherwise unparsed COM/CAUS suffix, and no unparsed syllables interrupt the feet. Hence, we can conclude that PCAO in Yidiɲ follows from the assumption that prelengthening suffixes are feet and general rules on foot structure: feet should be left-aligned and binary. I would like to emphasize that I do not assume that prelengthening suffixes come with length, i.e. in the form of a floating mora. Rather, the foot structure is the phonological element that sets them apart from other suffixes. Length on the preceding syllable is achieved by an additional rule that lengthens vowels before all syllables which are not part of a binary foot. This will be discussed in section 6.

(15) a. magi-(ri)-ŋa-l
[(ma.gi:).(ɻi).ŋal]
climb.up-GOING-COM/CAUS-CL
b. *[(ma.gi).ŋal.(ɻi)]
c. maɖinda-ŋa-(li)
[(ma.ɖin).(da.ŋa:).(li)]
walk.up-COM/CAUS-GOING
d. *[(ma.ɖin).da.(li).ŋa]

We have now seen how phonologically conditioned affix order in Yidiɲ can be explained from assumptions on foot structure. However, we have not discussed why there is no reordering with inflectional affixes in (8). In the following, I will argue that the affix asymmetries in final syllable deletion provide evidence for a stratal structure of the word, which in turn derives the absence of PCAO with inflectional affixes.

5 A formal analysis of final syllable shortening: evidence for strata

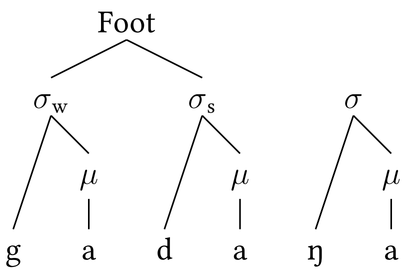
In the previous section, I have suggested that PCAO involves syllabification of roots and derivational affixes. I will now exploit this assumption to explain why roots and derivational affixes behave differently than inflectional affixes in final syllable deletion. Recall from section 3 that consonants of roots and derivational affixes are never deleted, while consonants of inflectional affixes may be deleted together with their vowels to produce licit codae. The relevant data is in section 3, the examples showing the blocking of consonant deletion is repeated in (16). Crucially, the COM/CAUS shortens if the resulting coda is allowed, as in (16-a). If shortening results in an illicit coda, as in (16-d), it is not possible to delete the entire affix (see (16-e)).

(16) Final shortening of COMIT/CAUS (Dixon 1977: 55, 349, 478)

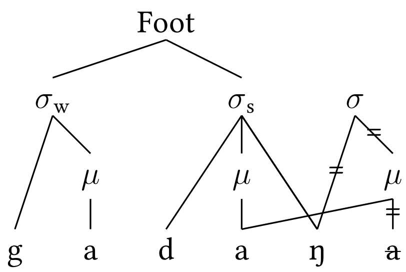
a. /gada-ŋa/
   [gada:ŋ]
   come-COM/CAUS
b. *[[gadaŋa]]
c. /guriɲ-ŋa/
   [(gu.ri:ɲ)ŋa]
   ‘good-COM/CAUS’
d. *[(gu.ri:ɲŋ)]
e. *[(gu.ri:ɲ)]

Let me first use the example in (16-a) to explain how final syllable deletion works. The illustration in (17) shows how the complex of root and COM/CAUS looks after syllabification has applied. We have already established that a first step of footing takes place, which builds binary foot from the left edge of the word. As a result, the first two syllables form a foot, while the third syllable, the COM/CAUS suffix, remains unparsed. In a later step, illustrated here in (18), final syllable deletion takes place. The process affects vowels of unparsed syllables, which are deleted. The former onset (*ŋ* in (18)) becomes the coda of the preceding syllable. Additionally, the mora formerly associated with the deleted vowel reassociates to the preceding vowel.

(17) First step: SYLLABIFICATION

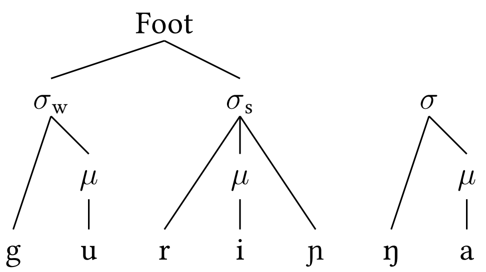


(18) Second step: FINAL SYLLABLE DELETION



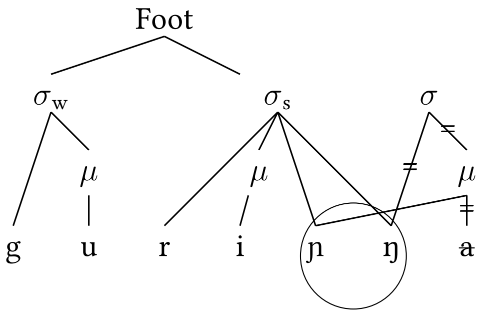
(16-c) presents a similar context with a bisyllabic root followed by the COM/CAUS suffix. Again, syllabification and footing applies from the left edge of the word and builds a binary foot from the root, leaving the COM/CAUS suffix unparsed, as illustrated in (19).

(19) First step: SYLLABIFICATION



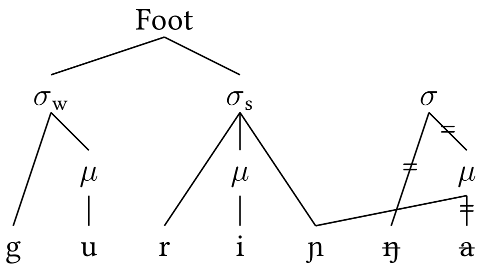
In contrast to the root *gada* in (16-a), the root *guriɲ* in (16-c) ends in a consonant. If we applied final syllable deletion as illustrated in (18), it would result in an illicit coda combination *ɲŋ*, as shown in (20).

(20) Second step: Final syllable deletion produces illicit coda



To avoid unparsed syllables, it seems conceivable to delete the syllable altogether. This option is sketched in (21). As shown in (16), this option does not surface either. Since deleting the entire syllable results in the deletion of the entire affix, one might think that the ungrammaticality of (21) results from a ban the deletion of entire affixes (similar to a Realize Morpheme constraint by Kurisu 2001). However, this reasoning seems unlikely given that we have seen the deletion of entire morphemes in (13-d). I assume that consonant deletion is blocked because the consonant is already part of a syllable. In short, syllabification protects consonants, but not vowels, from deletion.

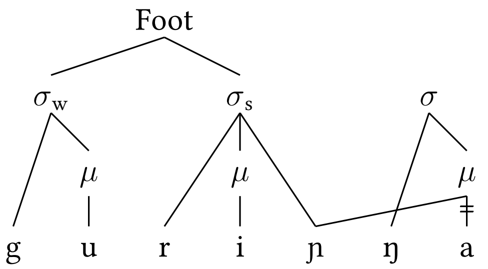
(21) Second step: Final syllable deletion with disallowed consonant deletion



The observed surface form is illustrated in (22). The example in (16-c) shows that the final vowel is not deleted, yet the preceding vowel lengthens. I assume that the final mora deassociates from the final vowel and reassociates with the preceding one, probably as a result from a ban against moras not dominated by a binary foot. The illustration in (22) reveals another parallel between PCAO and final syllable deletion: vowel lengthening occurs before syllables which are not part of a binary foot. In the previous section, we have shown that a vowel is lengthened before unary feet in form of the prelengthening suffixes. In final syllable deletion, lengthening occurs before unparsed moras, whose syllable may be deleted (see (18)) or not, as in (22).[2]

[2]It is also conceivable that the moras are actually linked to both vowels, rather than de- and reassociated. It is hard to find empirical arguments for either option. On the theoretical level, the former option would create syllables with mora-less vowels while the latter creates moras linked to two feet at the same time. I adopt the former option in this paper, but I truly believe the latter option would be compatible with all other assumptions I make in this paper.

(22) Second step: No Final syllable deletion, only mora reassociation



I have now developed a formal description of PCAO and final syllable deletion, yet we have not explained why inflectional affixes do not participate in PCAO, nor why their consonants may shorten along with their vowels. The relevant data is repeated in (23). Crucially, (23-b) is ungrammatical due to the illicit coda cluster. Instead, (23-a) surfaces, where the /*d*/ has been deleted.

(23) Final syllable shortening in inflectional affixes (Dixon 1977: 277)

a. /binarŋa-l-ɲunda/
[(bi.nar)(ŋalɲu:n)]
'warn-CL-DAT.SUBORD'

b. *[(bi.nar)(ŋalɲu:nd)]

c. *[(bi.nar)(ŋalɲu:n).da)]

d. /wawa-l-ɲu/
[wawa:l]
'see-CL:L-PST'

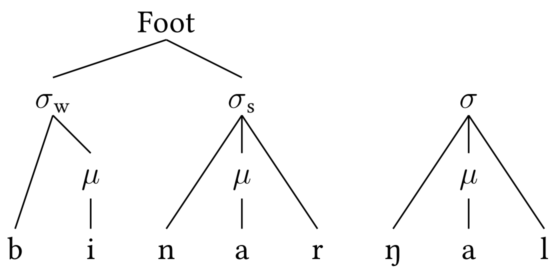
We have already said that roots and derivational affixes undergo syllabification and footing. (23) does not contain any derivational affixes, only a trisyllabic root *binarŋa* and the inflectional suffix *ɲu-nda*. This leads us to another crucial assumption of the analysis I lay out in this paper. I assume that the asymmetries between derivational and inflectional affixes in PCAO and final syllable deletion arises from the fact that they belong to two different morphophonological domains. The root and the derivational affixes belong to the *stem level*, where PCAO and syllabification apply. Inflectional affixes, in contrast, are always positioned to the right of the derivational affixes. I assume that inflectional affixes belong to a different domain, the *word level*, where final syllable deletion takes place and syllabification applies again. This idea neatly derives both asymmetries between derivational and inflectional affixes: inflectional affixes do not participate in PCAO because they have not been concatenated when PCAO takes place. Consonants of derivational affixes are protected by syllabification because they had already been syllabified at a previous step of phonological optimization. We can summarize the cyclic architecture of the word in Yidiɲ in table 2.

morphology: phonology:	concatenation of root and derivational affixes PCAO & syllabification	*stem level*
	bracket erasure	
morphology: phonology:	concatenation of inflectional affixes final syllable deletion & syllabification	*word level*
	bracket erasure	

Table 2: Assumed architecture of the morpho-phonology interface

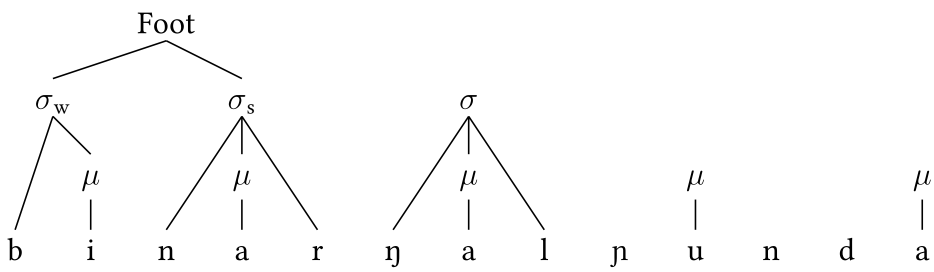
I assume that morphology and phonology are interleaved. Morphology builds words by concatenating roots and affixes. Afterwards, phonological rules apply and manipulate the output of morphology. Crucially, phonology has access to morpheme boundaries, i.e. phonologically conditioned affix order is phonologically optimizing reordering of morphemes. After each level, bracket erasure takes place, which renders morphological structure inaccessible to further cycles. Bracket erasure is a mechanism introduced by Kiparsky (1982a,b) and refers to the process of making morphological boundaries invisible to phonological or morphological rules. More specifically, morphological boundaries become invisible at the end of a cycle. Consequently, neither phonological nor morphological rules can make reference to these boundaries.
Let me now explain how this cyclic architecture accounts for the deletion of consonants of inflectional affixes in final syllable deletion. Given that (23-a) does not have any derivational affixes, only the root becomes syllabified at the stem level. Again, we use the first two syllables to build a binary foot at the left edge of the word, leaving the third syllable unparsed. This is illustrated in (24).

(24) Stem level: Syllabification



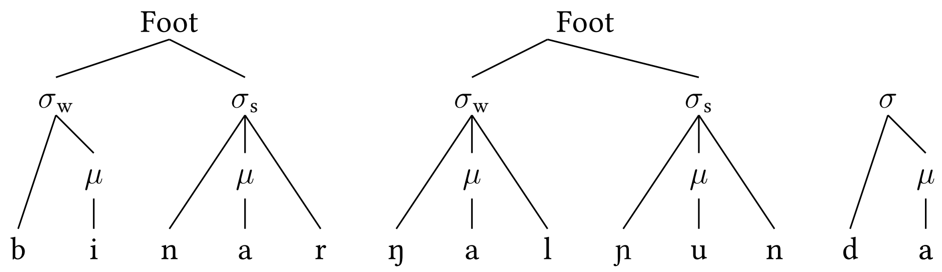
At the next level, the *word level*, the inflectional affixes are concatenated. While the root is now parsed into syllables, the inflectional affix *ɲu-nda* is not. This step is demonstrated in (25).

(25) Word level: Concatenation of inflectional affixes



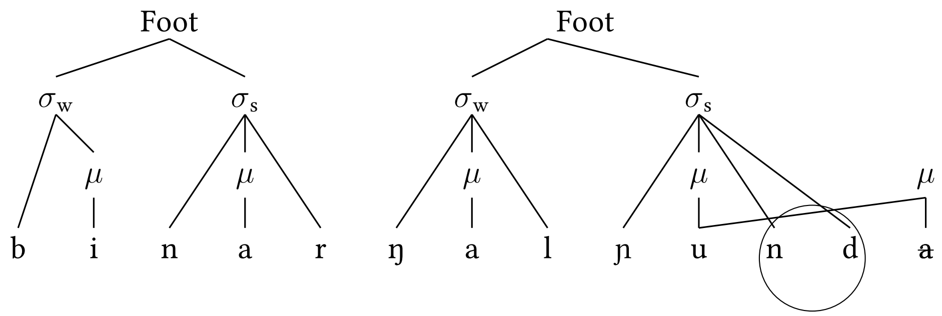
After the morphological component of the *word level*, phonological optimization applies. Just as at *stem level*, the grammar is trying to produce binary feet from the left. The unparsed syllable of the root is parsed into a foot with the first syllable of the inflectional affix /*ɲun*/, while the second syllable of the affix /*da*/ remains unparsed. In contrast to the *stem level*, unparsed syllables are not tolerated, causing the structure in (26) to become ungrammatical.

(26) Word level: Without Final syllable deletion



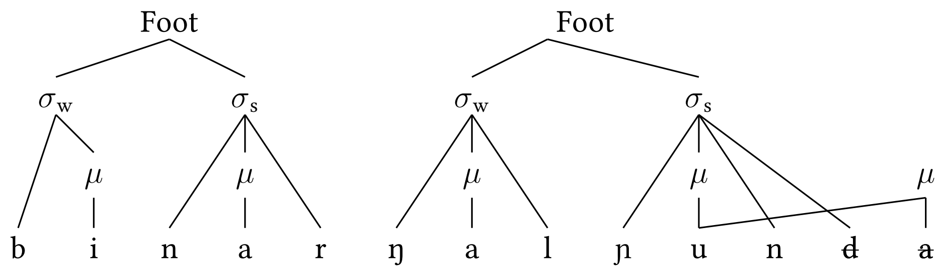
The ban against unparsed syllables causes final syllable deletion. If we delete the final vowel and reassociate the former onset to the coda of the preceding syllable, an illicit coda combination /*nd*/ arises, and (27) cannot surface.

(27) Word level: Final syllable deletion without consonant deletion



In the examples with derivational affixes above, I have shown that illicit coda combinations prevent final syllable shortening from applying. With inflectional affixes, however, we observe that final syllable deletion applies nonetheless, deleting the onset consonant along with the final vowel. This can abstractly be visualized by the illustration in (28). Crucially, the entire final syllable is deleted, with only its mora being retaining by reassociating to the preceding vowel. The important difference between derivational and inflectional affixes is that derivational affixes have already undergone syllabification at stem level, while inflectional affixes have not. Thus, deleting the consonant of derivational affixes affects syllabic structure, while deleting the consonant of inflectional affixes affects only the consonant itself.

(28) Word level: Final syllable deletion with consonant deletion



In the previous two sections, I have combined two important assumptions which together explain the asymmetries between derivational and inflectional affixes in phonologically conditioned affix order and final syllable deletion. Importantly, I assume that prelengthening suffixes do not come with floating moras but with foot structure, and secondly, I assume that the word in Yidiɲ is layered. Derivational affixes belong the first layer, the stem level, while inflectional affixes belong to the second layer, the word level. At stem level, phonological rules affect the order of morphemes to create an optimal foot structure. At word level, unparsed syllables are shortened. Evidence for the cyclic architecture of the word comes from asymmetries between derivational and inflectional affixes in final syllable deletion. Consonants of derivational affixes are protected by syllabic structure, which suggests that these affixes have already undergone a cycle of phonological optimization. Inflectional affixes, on the other hand, have not been subject to syllabification, hence their consonants are not protected. In the following, I will transfer these assumptions into an analysis in Stratal Optimality Theory.

6 A StratOT analysis of PCAO

The assumptions I have made so far lend themselves to a cyclic model with two different blocks of phonological rules: a first rule block that triggers PCAO at stem level and a second rule block that enforces final syllable shortening and also vowel lengthening at the word level. I have also shown that PCAO in Yidiɲ is phonologically optimizing. The tight interaction between morphology and phonology requires a theoretical model in which phonological rules have access to morpheme boundaries. It is argued most prominently by Embick (2010) that a global interaction of phonology and morphology should be banned. One reason for this objection comes from Paster (2006, 2009), who argues that a model where phonology has unlimited access to morphological structure would overgenerate by predicting unrestricted non-local phonological requirements on morphology. In cyclic models, however, the interaction of phonology and morphology is temporarily limited to cyclic domains and therefore less powerful, as demonstrated by Orgun & Inkelas (2002), Deal & Wolf (2017) or Benz (2017), among others.
Within this analytical space, Stratal Optimality Theory (StratOT) (Kiparsky 2000, Bermúdez-Otero 2011, 2018) has been suggested as a derivational version of Standard Parallel Optimality Theory (Prince & Smolensky 1993). StratOT is based on assumptions similar to the ones posited by Lexical Phonology and Morphology (Kiparsky 1982a) and includes the division of labor into several different cyclic domains. A concrete suggestion concerning the number of domains comes from Bermúdez-Otero (2011), who assumes three different levels, and crucially, exactly two word-internal strata:[3] the *stem level* comprises the root and some derivational affixes, the *word level* includes the stem and inflectional affixes, and the *phrase level* comprises entire utterances and more external clitic-like affixes.
Another crucial component of StratOT is re-ranking. This assumption explains the generalization that certain phonological rules such as PCAO apply only to certain subdomains, suggesting that the ranking of the constraints may differ from one stratum to the other. Within one stratum, the phonological grammar is regulated by the interaction of universal and violable constraints, just as in SPOT. Let us now see how this model explains the affix asymmetries and processes in Yidiɲ. We have already said PCAO takes place that the first

[3] The classification by Bermúdez-Otero (2011) is thought to serve only as a very rough categorization of affixes. Reports and analyses from individual languages suggest that there are languages with more than three cyclic domains (see Hargus 2018 on Sekani, Rice 1989 on Slave, Odden 1996 for Kimatuumbi, Caballero 2008 for Choguita Rarámuri, Eberhard 2009 for Mamaindê, Jones 2014 for Kinande or Jaker & Kiparsky 2020 for Tetsǫ̣t'ıné). Moreover, post-lexical cyclic domains containing clitics or phrases are described by Kaisse (1985, 1990), Clark (1990), McHugh (1990), Rubach (2011, 2016), Jones (2014) or Gjersøe (2016). A priori, it is not clear, how many rule blocks can be established within a language or across languages. Moreover, Booij (1996) demonstrates that inflection should be differentiated further, therefore suggesting to distinguish between contextual inflection (e.g. agreement) and inherent inflection (e.g. tense, mood, aspect).

cycle, the stem level, and arises from the interaction of different rules on the wellformedness of feet. Concretely, we have reached the generalization that prelengthening affixes are feet, which should be left-aligned. This explains why (29-a) is better than (29-b), which has an unparsed syllable between the footed GOING suffix and the left edge of the word. Moreover, this explains why (29-c) is more optimal and (29-d), where an unparsed syllable intervenes between the footed affixes and the root. Data is repeated from (15).

(29) a. magi-(ri)-ŋa-l
[(ma.giː).(ɻi).ŋal]
climb.up-GOING-COM/CAUS-CL
b. *[(ma.gi).ŋal.(ɻi)]
c. maɖinda-ŋa-(li)
[(ma.ɖin).(da.ŋaː).(li)]
walk.up-COM/CAUS-GOING
d. *[(ma.ɖin).da.(li).ŋa]

From these generalizations, I assume the constraints in (30). Left alignment of feet is regulated by the constraint ALIGN(FT,L,PRWD,L), which penalizes every syllable intervening between the left edge and the left edge of each foot. DEP_{FOOT} is the highest ranked constraint and practically inviolable. It ensures that previously created or existing foot structure cannot be manipulated. The lowest ranked constraint is LINEARITY-IO, a faithfulness constraint that penalizes manipulated morpheme order. Two more constraints are relevant at the stem level. PARSESYLLABLE penalizes syllables that are not part of a foot in the output. However, parsing syllables into a foot is only possible if this does not violate FOOTBINARITY. The exact definition of FOOTBINARITY I adopt refers to mora. Crucially, the constraint penalizes each mora which is not part of a bisyllabic foot.

(30) List of constraints relevant at stem level

1. ALIGN(FT,L,PRWD,L): (McCarthy & Prince 1993a,b)
   The left edge of every foot coincides with the left edge of the prosodic word. Assign a * for each syllable intervening between misaligned edges.
2. FOOTBINARITY:
   Assign a * for each μ which is not part of a binary foot, i.e. a foot head dominating two syllable heads.
3. DEP_{FOOT}:
   Assign a * for each association line between a foot head and a syllable head in the output which has no correspondent in the input.
4. PARSESYLLABLE:
   Assign one * for each vowel in the input that is not dominated by a Foot head in the output.
5. LINEARITY-IO:
   Assign a * for each morpheme A which precedes morpheme B in the input, iff A and B have correspondents in the output and the correspondent of A follows the correspondent of B in the output.

The interplay of these constraints at stem level is visualized in the tableaux in (31) and (32). Recall that I assume an independent cycle of morphological optimization which applies prior to the phonological optimization at stem level. Morphology creates the input to phonology by concatenating the affixes of the stem level. I give a hypothetical input in both (31) and (32). However, I would like to clarify that we have no clue which order is actually underlying (and hence, which order causes a violation of Linearity-IO). Based on general assumptions on affix order, we might have taken the interpretation of the forms as a clue to the semantic composition of the affixes, which can be assumed to be a driving factor of affix order (Baker 1985, Rice 2000). However, Dixon (1977) does not provide a translation of the forms, so I see no way to reveal the underlying order. I give opposite underlying orders in (31) and (32) (com/caus before going in (31) and going before com/caus in (32)) to illustrate that the unrecoverable underlying order does not play a role: since Linearity-IO is the lowest ranked constraint, it does not matter if it is violated or not. The observed surface form is seen in candidate a in (31). The footed going suffix is shifted to the position to the right of the root to minimise violations of Align(Ft,L,PrWd,L). We see that this affix still creates a violation of FootBinarity. However, we cannot merge it with the unparsed com/caus affix, since this would cause a fatal violation of Dep$_{\text{Foot}}$, as in candidate d. Not shifting the affix would result in more violations of Align(Ft,L,PrWd,L), independent of whether the preceding material consist of a binary foot and an unparsed syllable, as in candidate b, or a tertiary foot as in candidate c. Note also that the tableau in (31) does not include the observed vowel length on the second syllable of the root. I assume that the lengthening rule applies only at word level, but makes reference to the foot structure built at stem level.

(31) Phonological optimization at stem level, (15-a)

/magi-ŋa-(ɻi)/	Dep$_{\text{Foot}}$	FootBinarity	ParseSyllable	Align(Ft,L,PrWd,L)	Linearity-IO
a. ☞ (ma.gi).(ɻi).ŋa		∗	∗	∗∗	∗
b. (ma.gi).ŋa.(ɻi)		∗	∗	∗∗∗!	
c. (ma.gi.ŋa).(ɻi)		∗∗!∗∗		∗∗∗	
d. (ma.gi).(ɻi.ŋa)	∗!			∗∗	

With a trisyllabic root, the same constraint profile creates a different surface order, as seen in (32). The trisyllabic root cannot form a foot on its own as in candidate e, since this creates too many violations of FootBinarity. However, building a binary foot of the first two syllables and merging the third root syllable with the going suffix manipulates existing foot structure, thus causing a fatal violation of Dep$_{\text{Foot}}$ in candidate d. Having the third syllable of the root build a foot on its own again causes a violation of FootBinarity in candidate c. Leaving the third syllable of root unparsed in candidate b avoids this violation of FootBinarity but causes a fatal violation of ParseSyllable. Thus, shifting the going affix to the right edge in candidate a becomes optimal despite causing a violation of the lowest ranked constraint Linearity-IO.

(32) Phonological optimization at stem level, (15-c)

/maɖinda-(li)-ŋa/		DEP$_{\text{Foot}}$	FootBinarity	ParseSyllable	Align(Ft,L,PrWd,L)	Linearity-IO
a. ☞	(ma.ɖin).(da.ŋa).(li)		*		******	*
b.	(ma.ɖin).da.(li).ŋa		*	*!*	***	
c.	(ma.ɖin).(da).(li).(ŋa)		**!*		*********	
d.	(ma.ɖin).(da.li).ŋa	*!		*	**	
e.	(ma.ɖin.da).(li).ŋa		**!**	*	***	

Once the phonological computation at stem level is complete, bracket erasure applies and deletes morpheme boundaries. At word level, morphology connects the complex of root and derivational affixes with the inflectional affixes. As mentioned earlier, I assume that final syllable deletion and lengthening apply in the phonological component of the word level. Crucially, I assume that both processes are triggered by one and the same constraint FootBinarity. FootBinarity penalizes moras that are not part of a binary foot. I assume that this constraint causes the mora to reassociate with its preceding vowel, which is then lengthened. The remaining mora-less vowel is deleted despite causing a violation of the faithfulness constraint Max, the general anti-deletion constraint. These vowels delete because they cannot survive without a mora, due to a violation of *NoMora, which penalizes mora-less vowels. Footed affixes like the going affix are unary feet, thus causing a violation of FootBinarity. However, their vowel is never deleted due to a more specific faithfulness constraint Max(V)$_{\text{Foot}}$, which protects footed vowels from deletion. Hence, footed structures survive deletion despite causing a violation of *NoMora. Another crucial interplay happens between the constraints LicitCoda, which penalizes illicit coda clusters, and Max(C)$_\sigma$, which protects syllabified consonants. Deleting the vowel to avoid violations of FootBinarity and *NoMora causes the onset consonant to become part of the preceding coda. If this results in an illicit coda, we might delete the consonants along with their vowels to not violate LicitCoda. However, this only happens to material of the word level, since stem level material has already been syllabified and is therefore protected by Max(C)$_\sigma$.

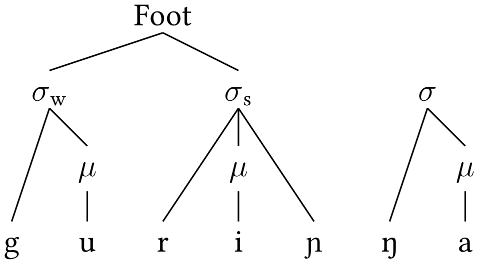
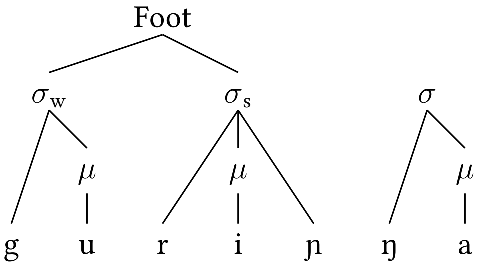
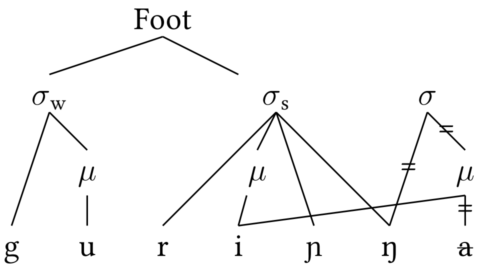
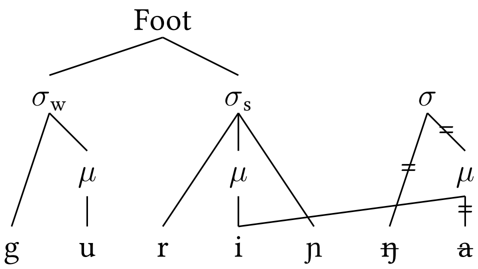
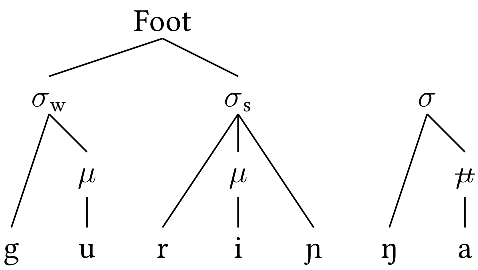
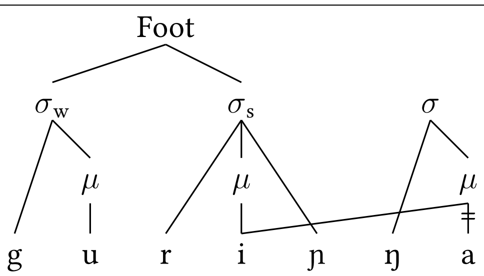
(33) List of constraints relevant at word level

1. FootBinarity:
   Assign a * for each μ which is not part of a binary foot, i.e. a foot head dominating two syllable heads.
2. Max(μ):
   Assign a * for each μ in the input which has no correspondent in the output.
3. Max(C)$_\sigma$:
   Assign a * for each consonant dominated by a syllable head in the input which has no correspondent in the output.
4. Max(V)$_{\text{Foot}}$:
   Assign a * for each vowel dominated by a foot head in the input which has no correspondent in the output.
5. *NoMora:
   Assign a * for each vowel which is not associated with a mora.

6. MAX:
   Assign a * for each segment in the input which has no correspondent in the output.
7. LICITCODA:
   Assign a * for each syllable which has an illicit coda.

Let me illustrate how the interaction of these constraints derives the asymmetries between derivational and inflectional affixes in final syllable deletion, as well as the prelengthening effect of footed affixes. The tableau in (34) illustrates how derivational affixes are affected by word level phonology. At this level, FOOTBINARITY penalizes the mora of the unparsed COM/CAUS affix in candidate a, which matches the stem level output. We cannot simply delete the mora, as in candidate d, since this causes a fatal violation of the constraint MAX(μ). Candidate b deletes the final vowel and reassociates its mora. However, this causes an illicit coda, which leads to a fatal violation of the highest ranked constraint LICITCODA. Candidate c avoids this violation by deleting the affix consonant along with the vowel. However, this creates a fatal violation of MAX(C)$_\sigma$, which penalizes the deletion of syllabified consonants. The only remaining option is to keep the vowel and the consonant while reassociating the mora to the preceding consonant. This creates a violation of *NOMORA, yet becomes optimal.

(34) Phonological optimization at word level, (12)

	LicitCoda	Max(V)$_{Foot}$	Max(C)$_{\sigma}$	Max(μ)	FootBinarity	*NoMora	Max
a. 					*!		
b. 	*!						*
c. 			*!				**
d. 				*!		*	
e. ☞ 						*	

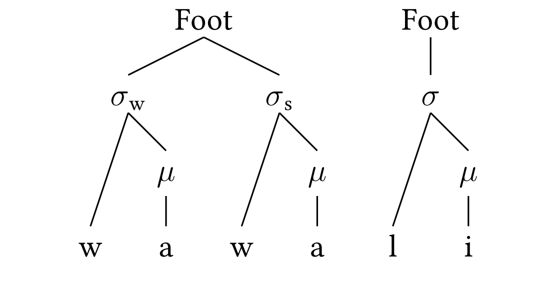
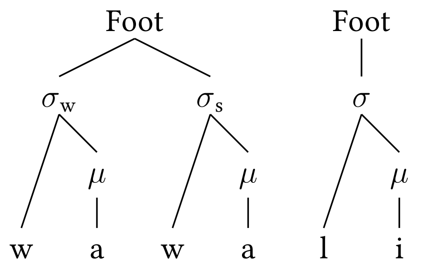
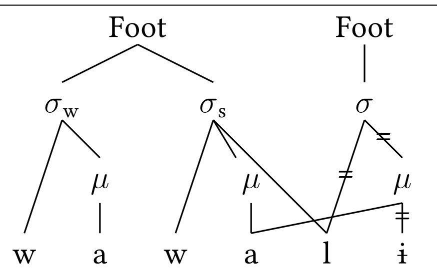
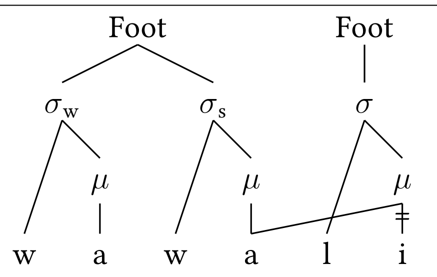
In contrast to derivational affixes, inflectional affixes have not been syllabified at stem level yet. When they enter the morphological derivation at word level, they do not come with syllable structure, as illustrated in the input of the tableau in (35). The phonological grammar again builds binary foot from the left to minimize violations of FootBinarity, see candidate a, which corresponds to the output of the morphology. If we take the two leftmost moras and merge them into a binary foot, as in candidate b, the final mora still causes a fatal violation of FootBinarity. Candidate e shows that deleting the mora is not an option due to Max(μ). As in the tableau in (34), deleting the vowel and reassociating its mora causes a violation of LicitCoda due to the illicit coda combination /nd/ in candidate c. In contrast to derivational affixes, however, deleting the consonant in candidate d to avoid this violation becomes optimal since the constraint Max(C)$_\sigma$ refers to existing syllable structure in the input, which does not exist for inflectional affixes. Reassociation of the mora without deletion in candidate f, which was the optimal output of tableau (34) is ruled out by *NoMora, which penalizes the mora-less vowel in (35).

(35) Phonological optimization at word level, (13)

b i n a r ŋ a l ɲ u n d a	LicitCoda	Max(V)$_{Foot}$	Max(C)$_\sigma$	Max(μ)	FootBinarity	*NoMora	Max
a. b i n a r ŋ a l ɲ u n d a					*!**		
b. b i n a r ŋ a l ɲ u n d a					*!		
c. b i n a r ŋ a l ɲ u n d ~~a~~	*!						*
d. ☞ b i n a r ŋ a l ɲ u n ~~d~~ ~~a~~							**
e. b i n a r ŋ a l ɲ u n d a				*!		*	
f. b i n a r ŋ a l ɲ u n d a						*!	

A major advantage of the constraint-based analysis I suggest is that FootBinarity causes both final syllable deletion and the prelengthening of footed affixes, thus unifying the two phenomena. Tableau (36) illustrates how the constraint creates the prelengthening effect. In contrast to the unfooted com/caus affix, the going suffix enters the derivation as a foot on its own. Nonetheless, its mora creates a violation of FootBinarity in candidate a. Although deletion of the vowel would not create an illicit coda, deleting the vowel affects existing foot structure, thus causing a violation of Max(V)$_{Foot}$ in candidate b. As a result, the only remaining option is to reassociate the mora and keep the (now mora-less) vowel in candidate c, which becomes optimal despite a violation of *NoMora.

(36) Lengthening of footed affixes, (6)

	LicitCoda	Max(V)$_{\text{Foot}}$	Max(C)$_\sigma$	Max(μ)	FootBinarity	*NoMora	Max
a. 					$*!$		
b. 		$*!$					$*$
c. ☞ 						$*$	

7 Discussion

7.1 Non-shortening inflectional affixes and roots

In the previous sections, I have shown that asymmetric behaviour of affixes with respect to PCAO and final syllable shortening can be explained by assuming that a) prelengthening affixes are feet, and b) there are two different cyclic domains within the word. We have also seen that these assumptions bring about a unified source of vowel length: lengthening applies before syllables which are not part of a binary foot, independent of whether this syllable is deleted by final syllable deletion or not. In section 5, I already mentioned that most trisyllabic roots are shortening, while some others are not. Dixon (1977) observes

115 trisyllabic roots for which shortenening is not prohibited on phonotactic grounds, 34 of those 115 roots (≈ 30%) resist shortening. Crucially, the distinction is idiosyncratic and cannot be predicted from the phonological shape of root. The relevant examples from (14) are repeated here in (37). Both roots are almost identical, yet only *gaɟara* shortens, whereas *guɟara* does not. Round (2017) further notes that the exceptionality of those roots might be connected to the observation that these are mostly loaned forms.

(37) Shortening vs. non-shortening roots (Round 2017: 60)

a. /gaɟara/
[(ga.ɟa:r)]
'possum.ABS'

b. /guɟara/
[(gu.ɟa:).ra]
'broom.ABS'

We have also encountered inflectional affixes that resist shortening. The inflectional PURP affix *na* does not shorten in (38-a), although shortening would be phonotactically permissible, see (38-b).

(38) Non-shortening of PURP *na*

a. /gali-na/
[(ga.li:)na]
go-PURP

b. *[(ga.li:n)]

So far, we have connected the non-shortening of prelengthening affixes to the assumption that these affixes are feet. This assumption leads to the question whether all non-shortening morphemes, including roots and inflectional affixes can be assume to be feet. Round (2017) discusses the exceptionality of morphemes in final syllable deletion in Yidiɲ arguing that the idiosyncrasy arises from a *phonological diacritic*, rather than from a morphological diacritic or a representational difference in the phonological structure of the morphemes. The observed data and the proposed analysis is compatible with both ideas. If we would follow the argument by Round (2017), I would assume that prelengthening affixes are representationally different than other affixes, while non-undergoing roots and inflectional affixes are exceptional due to a phonological diacritic. If we were to assume that all non-undergoing affixes are feet, we would predict that shortening and non-shortening roots should behave differently with respect to PCAO. Crucially, PCAO should be blocked by non-shortening roots given that they come with unmanipulable foot structure. Unfortunately, this prediction cannot be tested at this point as Dixon (1977) does not provide a full list of non-shortening roots. Thus, I leave this question up for further research.

7.2 Lengthening after trisyllabic roots

As for vowel length, I have stated that it results from an interaction of the constraints FOOTBINARITY, which penalizes moras in footless syllables and unary feet, and MAX(μ), which prohibits the deletion of exactly those moras. As a result, those moras will reassociate to their preceding syllables. What does my analysis predict for contexts in which two or more consecutive syllables cause violations of FOOTBINARITY? This scenario arises if an unparsed syllable is followed by a prelengthening affix, as in (39), where a trisyllabic root takes a prelengthening affix. The analysis I put forward in section 6, predicts that the resulting foot structure should be *(wu.ŋa).ba.(ḍi)*, with both the unparsed syllable *ba* and the unary foot (*ḍi*) causing violations of FootBinarity. In these examples (and also in other cases of two FootBinarity-violating syllables), we observe that vowel length always appears on the first of those two, which is *ba* in (39-a). However, we would have predicted the last vowel of the preceding binary foot *(wu.ŋa)* to host the moras of both *ba* and the unary foot (*ḍi*), as in (39-b). There are, however, reasonable explanations why (39-b) does not surface, e.g. due to a constraint against trimoraic vowels (as in Gleim 2024), which has previously been claimed to hold universally by Bye (2005). We might also expect that each mora associates to its left, as in (39-c), i.e. the mora of *ba* associates to *ŋa*, while the mora of (*ḍi*) reassociates to *ba*. This form minimizes the violations of FootBinarity, but causes a clash of two long syllables, which is reported to be prohibited in other languages, as well, such as Slovak. Deleting the second long syllable to resolve this clash in (39-d) cannot become optimal, as it causes a violation of Max(μ). Since options (39-b-d) are suboptimal for independent reasons, the languages has no chance to avoid a double violation of FootBinarity. In this regard, the surfacing form in (39-a), where (*ḍi*) is moraless but *ba* has two moras, and the word level output *(wu.ŋa).ba.(ḍi)*, where both *ba* and (*ḍi*) have exactly one mora, are equally good with respect to FootBinarity. I see two possible explanations why the former option wins over the latter. First, we might assume that the unparsed syllable is not allowed to form a bimoraic foot on its own, thus avoiding a violation of ParseSyllable. Second, there could be another higher-ranked and more specific version of FootBinarity, which penalizes moras of unary feet in final position. All in all, the examples are clearly explicable with the assumptions about foot structure I have made so far.

(39) Trisyllabic roots and prelengthening affixes
  a. wuŋaba-ːḍi
     [(wu.ŋa).baː.(ḍi)]
     hunt.for-intrans
  b. *[(wu.ŋaːː).ba.(ḍi)]
  c. *[(wu.ŋaː).baː.(ḍi)]
  d. *[(wu.ŋaː).ba.(ḍi)]

7.3 The relative order of com/caus and intrans

In section 2, all examples of phonologically conditioned affix order involve the com/caus and the aspectual going affix. However, we have not discussed examples where the intrans suffix interacts with the com/caus suffix. The interaction of those affixes is particularly interesting as the functions of these affixes suggests a semantic interaction, as well. Previous work on affix order has clearly identified that the semantic composition is a driving factor in the linearization of affixes (Baker 1985, Rice 2000, Stiebels 2003, among many others), and relevant examples typically discuss the relative order of valency morphology. Let us first take a look at the examples in (40). (40-a) shows base verb *wawa* without any derivational affixes. In this sentence, the agent argument appears in ergative case, the instrument in instrumental case while the patient is in accusative case. From the distribution of the case markers, we can conclude that the base verb *wawa* is transitive. In (40-b), this base verb takes the intrans suffix, which roughly acts as an antipassive in this case. The formerly ergative marked agent now is absolutive marked in (40-b) whereas the formerly absolutive marked patient now takes oblique dative case. In (40-c), the COM/CAUS suffix appears to the right of the INTRANS suffix. It has the function of a comitative applicative: the instrument argument is now marked with structural absolutive case and the agent argument takes ergative case again, showing that the predicate is now transitive. Thus, the order of the argument follows the semantic composition. Since the COM/CAUS can only be added to intransitive bases, the INTRANS instantiates a prerequisite for the transitive root *wawa* to take a comitative applicative. The order of the affixes is also phonologically well-formed. The disyllabic root forms a foot on its own, just as the INTRANS suffix. Both feet are left-aligned and length appears before the unary foot, as predicted by the analysis.

(40) Intransitivizer preceding COM/CAUS (Dixon 1977: 320)

a. bama-:l waŋal-ḍi-ŋgu ŋaɲaɲ wawa:l
person-ERG boomerang-INST-ERG 1SG.ACC see.PST

b. bama waŋa:l-ḍi ŋanda wawa:-ḍi-ɲu
person.ABS boomerang-INST 1SG.DAT see-INTRANS-PST

c. bama-:l waŋal ŋanda wawa:-ḍi-ŋa:l
person-ERG boomerang.ABS 1SG.DAT see-INTRANS-COM/CAUS.PST
'The person with a boomerang saw me.'

In the example in (41-a), the COM/CAUS precedes the causative. The exact semantic effects of the affixes are harder to reveal than in the previous examples. *giwa* is listed as an intransitive root with a meaning roughly corresponding to English 'stir up'. With a COM/CAUS, the resulting base is transitive and takes the meaning 'tickle'. If we add the INTRANS suffix to this complex, the INTRANS takes an intransitivizing reflexive function. The relative order of the affixes is semantically transparent, as the reflexive function requires a transitive base. In this case, the resulting complex is not phonologically well-formed. Just as in (40-c), the complex consists of a bisyllabic root, an unfooted COM/CAUS and the footed INTRANS suffix. Hence, we would expect the same surface order despite having a different underlying semantic representation, which is only relevant in the morphological component of grammar. In short, we would expect (41-b) to surface overwriting the order suggested by the semantics. However, we see that PCAO fails to apply in (41-a). Rather, it seems that semantics prevent PCAO from applying. The simultaneous interaction of semantics and phonology is excluded in most models of grammar I am aware of, including the one I follow in this paper. A tentative explanation is that the contexts only seem to be phonologically comparable but differ in their phonological representations. Camacho Rios & Tallman (2024) give a recent and relevant discussion of the causative suffix in Southern Bolivian Quechua, which seems to appear in two different positions. They argue that the variable position derives from lexical causatives, where the causative is actually part of the root, and actual causative affixes, which are productively affixed to roots. Although the empirical facts do not allow me to apply the tests suggested by Camacho Rios & Tallman (2024), similar considerations might explain the pattern in (41-a). It could be the case that *giwaŋa* 'to tickle' is just stored as a whole unit, which is a foot on its own. Another possibility could be that causative *ŋa* is also a foot, while comitative *ŋa* is not. Both hypotheses make clear empirical predictions, however, the data situation does not allow me to test them, mainly because I do not have interpretations of the forms exhibiting PCAO. Hence, I have to leave this question open for further research.

(41) COM/CAUS preceding intransitivizer (Dixon 1977: 321)

a. ŋayu ganagayuy murŋgal-da giwa-ŋa-(ḍi)-ɲu
[ŋayu ganagayuy murŋga:l-da (gi.wa).ŋa.(ḍi:)ɲu]
1SG.ABS self feather-DAT stir.up-COM/CAUS-INTRANS-PST
'I tickled myself with a feather.'

b. *(gi.wa:).(ḍi).ŋaɲ

8 Conclusion

In this paper, I have analysed the non-uniform behaviour of different affixes with respect to two morphophonological phenomena. First, the relative order of the prelengthening aspectual affixes and the derivational COM/CAUS suffix *ŋa* is conditioned by the syllable structure of the root, whereas the order of the prelengthening aspectual suffixes and the phonologically similar inflectional affix *na* is not. Second, some morphemes are shortened in final position if deletion of the final vowel creates words with an even number of syllables. Crucially, prelengthening affixes never shorten. The COM/CAUS suffix *ŋa* and most inflectional affixes shorten, but resist shortening altogether if deletion results in an illicit coda. An important asymmetry between the COM/CAUS suffix and inflectional shortening affixes is that consonants of inflectional affixes may be deleted along with their vowels to produce licit codae, while the consonant of the COM/CAUS suffix never deletes. I argue that the asymmetries between the different types of affixes follow neatly from two major assumptions. First, I assume that prelengthening affixes are feet. Feet in Yidiɲ cannot be manipulated, hence resist final syllable deletion. The phonologically conditioned metathesis of affixes is driven by general rules on foot structure and foot placement, such as rules on foot binarity and the left alignment of feet. Second, I conclude that the word contains exactly two cyclic domains. Since PCAO takes place in the first domain, inflectional affixes cannot participate since they enter the word only at the second domain. Evidence for this claim comes from the asymmetry between the COM/CAUS suffix and inflectional suffixes in final syllable deletion – the COM/CAUS suffix has already been syllabified at the first cycle, thus preventing its consonant from deletion. I argue that these assumptions lend themselves to an analysis in Stratal Optimality Theory, which I demonstrate in section 6. The paper makes several important theoretical contributions. First, it provides a unified analysis of PCAO, final syllable deletion, and vowel length in Yidiɲ. Second, it provides additional support to the model of the morphology-phonology interface as predicted by Bermúdez-Otero (2018): the model assumes temporary access of the phonology to morpheme boundaries that predicting PCAO to apply **within one stratum**.

Abbreviations

ABS = absolutive, ACC = accusative, CAUS = causative, COM = comitative, DAT = dative, ERG = ergative, GOING = going aspect, CL = inflectional class, IMP = imperative, INTRANS = intransitivizer, NOM = nominative, NPST = nonpast, PRS = present, PST = past, PURP = purposive, SUBORD = subordinate

References

Baker, Mark. 1985. The Mirror Principle and morphosyntactic explanation. *Linguistic Inquiry* 16(3). 373–415.

Benz, Johanna. 2017. Stress and a morpheme ordering anomaly in Washo. Leipzig University BA thesis.

Bermúdez-Otero, Ricardo. 2011. The architecture of grammar and the division of labour in exponence. In Trommer, Jochen (ed.), *The morphology and phonology of exponence*, vol. 41, 8–83. Oxford: Oxford University Press.

Bermúdez-Otero, Ricardo. 2018. Stratal phonology. In *The Routledge handbook of phonological theory*, 100–134. London: Routledge.

Booij, Geert. 1996. Inherent versus contextual inflection and the split morphology hypothesis. In Booij, Geert & van Marle, Jaap (eds.), *Yearbook of morphology 1995*, 1–16. Dordrecht: Springer.

Bye, Patrik. 2005. Coda maximisation in Northwest Saamic. *Nordic Journal of Linguistics* 28(2). 189–221.

Caballero, Gabriela. 2008. *Choguita Rarámuri (Tarahumara) phonology and morphology*. Berkeley: University of California dissertation.

Camacho Rios, Glady & Tallman, Adam J. R. 2024. Word structure and constituency in Uma Piwra South Bolivian Quechua. In *Constituency and convergence in the Americas*, 603––646. Berli: Language Science Press. doi:10.5281/zenodo.10559861

Clark, Mary M. 1990. *The tonal system of Igbo*. Dordrecht: Foris.

Deal, Amy Rose & Wolf, Matthew. 2017. Outwards-sensitive phonologically-conditioned allomorphy in Nez Perce. In Gribanova, Vera & Shih, Stephanie (eds.), *The morphosyntax-phonology connection*, 29–60. Oxford: Oxford University Press.

Dixon, Robert MW. 1977. *A grammar of yidin*. Cambridge University Press.

Eberhard, David. 2009. *Mamaindê grammar: A Northern Nambikwara language and its cultural context*. Amsterdam: University of Amsterdam dissertation.

Embick, David. 2010. *Localism versus globalism in morphology and phonology*, vol. 60. Cambridge: MIT Press.

Gjersøe, Siri. 2016. Cyclic optimization of floating L tones in Kikuyu. *Available as ROA-1284 from the Rutgers Optimality Archive* .

Gleim, Daniel Christoph. 2024. *Countercyclic process interactions*: Leipzig University dissertation.

Hargus, Sharon. 2018. *The lexical phonology of Sekani*, vol. 6. New York: Garland.

Jaker, Alessandro & Kiparsky, Paul. 2020. Level ordering and opacity in Tetsǫ́t'ıné : a Stratal OT account. *Phonology* 37(4). 617–655.

Jones, Patrick Jackson. 2014. *Tonal interaction in Kinande: Cyclicity, opacity, and morphosyntactic structure*. Cambridge, MA: MIT dissertation.

Kaisse, Ellen M. 1985. *Connected speech: the interaction of syntax and phonology*. New York: Academic Press.

Kaisse, Ellen M. 1990. Toward a typology of postlexical rules. In Inkelas, Sharon & Zec, Draga (eds.), *The phonology-syntax connection*, 127–143. Chicago: University of Chicago Press.

Kiparsky, Paul. 1982a. Lexical phonology and morphology. In of Korea, The Linguistic Society (ed.), *Linguistics in the morning calm*, 3–91. Seoul: Hanshin.

Kiparsky, Paul. 1982b. Word-formation and the lexicon. In Ingemann, Frances (ed.), *1982 Mid-America linguistics conference papers*, 1–29. Lawrence: Department of Linguistics, University of Kansas.

Kiparsky, Paul. 2000. Opacity and cyclicity. *The Linguistic Review* 17(2-4). 351–366.

Kurisu, Kazutaka. 2001. *The phonology of morpheme realization*: University of California, Santa Cruz dissertation.

McCarthy, John J & Prince, Alan. 1993a. Generalized alignment. In Booij, Geert & van Marle, Jaap (eds.), *Yearbook of morphology 1993*, 79–153. Dordrecht: Springer.

McCarthy, John J & Prince, Alan. 1993b. Prosodic morphology: Constraint interaction and satisfaction .

McHugh, Brian. 1990. The phrasal cycle in Kivunjo Chaga tonology. In Inkelas, Sharon & Zec, Draga (eds.), *The phonology-syntax connection*, 217–242. Chicago: University of Chicago Press.

Odden, David. 1996. *The phonology and morphology of kimatuumbi*. Oxford: Clarendon.

Orgun, Cemil Orhan & Inkelas, Sharon. 2002. Reconsidering bracket erasure. In Booij, Geert & van Marle, Jaap (eds.), *Yearbook of morphology 2001*, 115–146. Dordrecht: Springer.

Paster, Mary. 2006. *Phonological conditions on affixation*. Berkeley: University of California dissertation.

Paster, Mary. 2009. Explaining phonological conditions on affixation: Evidence from suppletive allomorphy and affix ordering. *Word structure* 2(1). 18–37.

Prince, Alan & Smolensky, Paul. 1993. Optimality Theory: Constraint interaction in generative grammar. Tech. rep. RuCCS-TR-2 at Rutgers University.

Rice, Keren. 1989. *A grammar of Slave*. Berlin/New York: Mouton de Gruyter.

Rice, Keren. 2000. *Morpheme order and semantic scope: Word formation in the Athapaskan verb* (Cambridge Studies in Linguistics). Cambridge: Cambridge University Press.

Round, Erich. 2017. Phonological exceptionality is localized to phonological elements: The argument from learnability and Yidiny word-final deletion. In Bowern, Claire & Horn, Laurence & Zanuttini, Raffaella (eds.), *On looking into words (and beyond): Structures, relations, analyses*, 59–97. Language Science Press Berlin.

Rubach, Jerzy. 2011. Syllabic repairs in Macedonian. *Lingua* 121(2). 237–268.

Rubach, Jerzy. 2016. Polish yers: Representation and analysis. *Journal of linguistics* 52(2). 421–466.

Stiebels, Barbara. 2003. Transparent, restricted and opaque affix orders. In Junghanns, Uwe & Szucich, Luka (eds.), *Syntactic structures and morphological information*, 283–315.